



**M. Kumarasamy
College of Engineering**

Autonomous | Affiliated to Anna University Chennai

Approved by AICTE, Accredited by NBA and NAAC
Thalavapalayam, Karur - 639 113, Tamilnadu, India.



A Project Report

on

ONLINE SHOPPING CART MANAGER

Submitted in partial fulfillment of requirements for the award of the course

of

CGB1121-PYTHON PROGRAMMING

Under the guidance of

Mrs. I. KARTHIKA M.E

Assistant Professor / CSE

Submitted By

KARTHIKA S 927625BCS068

NETHRA V 927625BCS105

DEPARTMENT OF FRESHMEN ENGINEERING

M.KUMARASAMY COLLEGE OF ENGINEERING

(Autonomous)

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BONAFIDE CERTIFICATE

This is to certify that this project report on “**ONLINE SHOPPING CART MANAGER**” is the bonafide work of **KARTHIKA S (927625BCS068)**, **NETHRA V(927625BCS105)** who carried out the project work during the academic year 2025 - 2026 under my supervision.

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Submitted for the End Semester Review held on _____.

INTERNAL EXAMINER

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VISION OF THE INSTITUTION

To emerge as a leader among the top institutions in the field of technical education

MISSION OF THE INSTITUTION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges
- Create a diverse, fully-engaged, learner-centric campus environment to provide quality education to the students
- Maintain mutually beneficial partnerships with our alumni, industry, and Professional associations

VISION OF THE DEPARTMENT

To achieve education and research excellence in computer Science and Engineering

MISSION OF THE DEPARTMENT

- To excel in academic through effective teaching learning techniques.
- To promote research in the area of computer science and engineering with the focus on innovation.
- To transform students into technically competent professionals with societal and ethical responsibilities.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- **PEO1:** Graduates will have successful Career in software industries and R&D divisions through continuous learning
- **PEO2:** Graduates will provide effective solutions for real world problems in the key domain of computer science and engineering and engage in lifelong learning.
- **PEO3:** Graduates will excel in their profession by being ethically and socially responsible

PROGRAM OUTCOMES (POs)

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
3. **Design/development of solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required..
4. **Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling recognizing their limitations to solve complex engineering problems.
6. **The engineer and the world:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
7. **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
8. **Individual and collaborative team work:** Function effectively as an individual, and as a member or leader in diverse/multidisciplinary teams.
9. **Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
10. **Project management and finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11.Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- **PSO1: Professional Skills:** Ability to apply the knowledge of computing techniques to design and develop computerized solutions for the problems.
- **PSO2: Successful career:** Ability to utilize the computing skills and ethical values in creating a successful career.

ABSTRACT

The Online Shopping Cart Manager is a Python-based application designed to simulate the core functionality of an e-commerce shopping cart system. This project uses lists and basic data structures to store and manage items selected by users. It allows users to add new products, remove existing items, update quantities, and view the cart contents efficiently . The system dynamically calculates the total cost based on the items and their quantities, ensuring accurate billing. By implementing fundamental programming concepts such as loops, conditional statements, and functions, this project demonstrates how real-world online shopping platforms manage cart operations. Overall, the project provides a simple yet effective simulation of an online shopping cart, enhancing understanding of data management and user interaction in Python applications.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	COs MAPPED	POs MAPPED	PSOs MAPPED
The Online Shopping Cart Manager is a Python-based application designed to simulate the core functionality of an e-commerce shopping cart system. This project uses lists and basic data structures to store and manage items selected by users. It allows users to add new products, remove existing items, update quantities, and view the cart contents efficiently . The system dynamically calculates the total cost based on the items and their quantities, ensuring accurate billing. By implementing fundamental programming concepts such as loops, conditional statements, and functions, this project demonstrates how real-world online shopping platforms manage cart operations. Overall, the project provides a simple yet effective simulation of an online shopping cart, enhancing understanding of data management and user interaction in Python applications.	CO1	PO1(3)	PSO1(2)
	CO2	PO2(3)	PSO2(3)
	CO3	PO3(3)	
	CO4	PO4(2)	
		PO5(2)	
		PO6(3)	
		PO7(2)	
		PO8(3)	
		PO9(2)	
		PO10(3)	
		PO11(3)	

Note: 1- Low, 2-Medium, 3- High

SUPERVISOR

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM STATEMENT

In today's digital world, online shopping has become very common, but managing a shopping cart efficiently is an important part of any e-commerce system. Many beginners face difficulty in understanding how shopping cart systems work internally, such as adding items, updating quantities, removing products, and calculating total cost accurately. The problem is to design and develop a simple system that simulates the basic functionalities of an online shopping cart. The system should allow users to manage their selected items easily and perform operations like adding, deleting, and updating products while ensuring correct billing. This project aims to solve this problem by creating a Python-based application that demonstrates how a shopping cart works using fundamental programming concepts like lists, loops, and conditional statements.

1.2 INTRODUCTION

Online shopping has become an essential part of modern life, allowing people to purchase products easily from anywhere at any time. A key component of any e-commerce platform is the shopping cart system, which helps users select, manage, and review the items they wish to buy before making a purchase. The Online Shopping Cart Manager is a Python-based project developed to simulate the basic functionalities of a real-world shopping cart. This system allows users to add products, remove items, update quantities, and view the contents of their cart in an organized manner. It also calculates the total cost based on the selected items and their quantities, ensuring accurate results. This project is designed using fundamental programming concepts such as lists, loops, conditional statements, and functions. It provides a clear understanding of how data is stored and managed in an application. The main aim of this project is to help learners understand the working process of a shopping cart system in a simple and effective way.

1.3 PYTHON PROGRAMMING CONCEPTS

This project uses basic Python programming concepts to implement the shopping cart system. Lists are used to store product names and prices. Conditional statements (if-else) help in making decisions such as adding or removing items. Loops are used to display products and repeat menu options. Functions are used to organize the code into small, reusable blocks. Variables are used to store user inputs and calculate the total cost. These concepts together help in building a simple and efficient shopping cart application.

1.4 OBJECTIVE

The main objectives of the Online Shopping Cart Manager project are:

- To develop a user-friendly application that simulates the basic functionalities of an online shopping system.
- To implement secure user registration and login functionality for maintaining user data safely.
- To design an efficient product management system that allows users to view and select products easily.
- To create a shopping cart module where users can add, update, and remove items as per their requirements.
- To provide real-time calculation of total cost based on selected products and quantities.
- To implement an order processing system that records transaction details such as items purchased, total amount, and date/time.
- To ensure data persistence by storing all user and transaction data using a database (SQLite).
- To apply core Python programming concepts such as GUI design, database connectivity, modular programming, and event-driven programming.
- To build a structured and maintainable system using a modular approach.

CHAPTER 2

PROJECT METHODOLOGY

2.1PROPOSED WORK

The proposed Online Shopping Cart Manager is designed to provide a simple, fast, and efficient online shopping experience for customers. The system enables users to browse different products, add selected items to the shopping cart, modify product quantities, remove unwanted items, and complete purchases through secure payment methods. It also allows administrators to manage product details, customer information, stock availability, and order records efficiently. The project helps reduce manual effort, minimizes errors in billing and order management, saves time, and improves customer satisfaction by offering a convenient and user-friendly shopping platform.

2.2 ARCHITECTURE

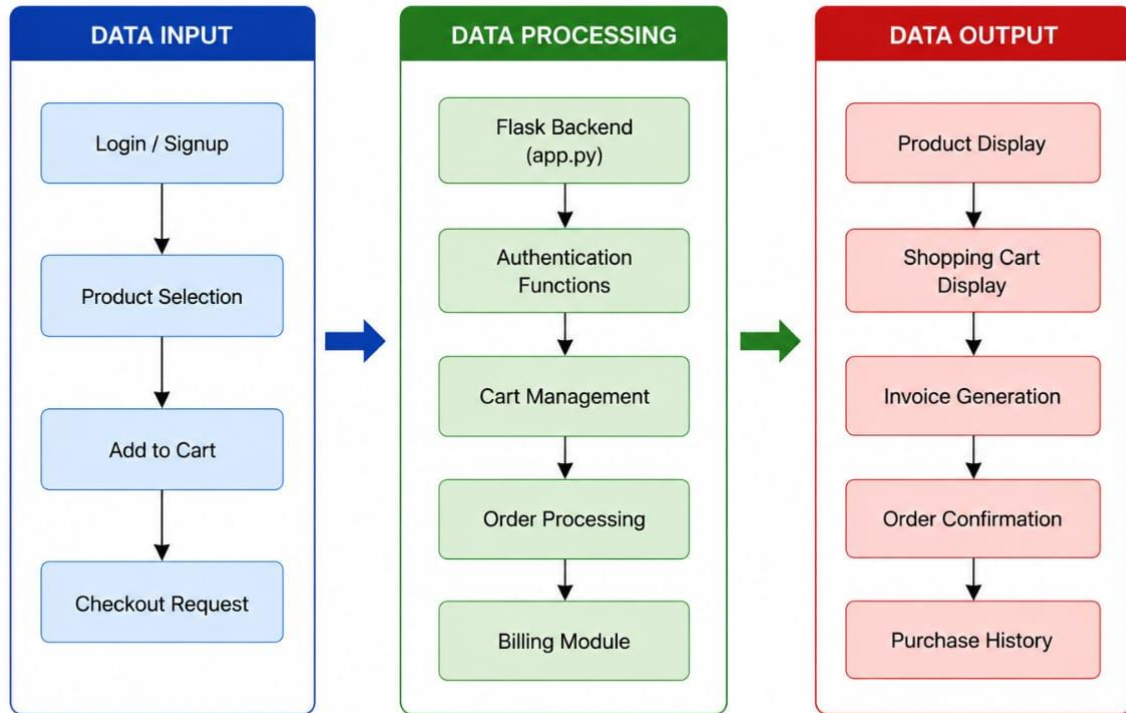


Fig 2.1 Architecture Diagram for Online Shopping Cart Manager

CHAPTER 3

SOFTWARE REQUIREMENTS

Python 3.14.4

It is a high-level programming language used to develop the backend of the Online Shopping Cart Manager application. Python is used for processing user requests, managing shopping cart operations, handling product details, and connecting the application with the database. It is simple, readable, and efficient for web application development.

VS Code

It is an Integrated Development Environment (IDE) used for writing, editing, and testing the project code. VS Code provides a user-friendly interface and supports debugging, code management, and project execution effectively.

GUI Framework

It is used to create an interactive and user-friendly graphical interface for the Online Shopping Cart Manager application. The GUI framework helps users easily browse products, add items to the cart, place orders, and navigate through the application smoothly. It improves the overall appearance and usability of the system.

CHAPTER 4

MODULE DESCRIPTION

4.1 User Interface Module

The User Interface Module is designed to provide a simple and user-friendly environment for customers to interact with the Online Shopping Cart Manager system. It allows users to easily browse products, search items, add products to the cart, and place orders smoothly through an attractive interface.

4.2 Product Management Module

The Product Management Module is used to manage product details within the system. It helps administrators add, update, delete, and maintain product information such as product name, category, price, and stock availability efficiently.

4.3 Cart Management Module

The Cart Management Module helps users manage selected products before purchasing. It allows customers to add items to the cart, update product quantities, remove unwanted items, and view the total price of selected products easily.

4.4 Billing Module

The Billing Module is used to generate bills and process payments for customer orders. It calculates the total amount, manages payment details, and provides accurate billing information for successful order completion.

4.5Display Module

The Display Module is responsible for showing product details, cart information, billing details, and order status to users. It helps in presenting information clearly and improves the overall user experience of the application.

CHAPTER 5

IMPLEMENTATION

5.1 CODING

```
import tkinter as tk

from tkinter import messagebox

import sqlite3

import datetime

import random

# ----- DATABASE -----

conn = sqlite3.connect("shopping.db", check_same_thread=False)

cur = conn.cursor()

cur.execute("""

CREATE TABLE IF NOT EXISTS users (

    id INTEGER PRIMARY KEY AUTOINCREMENT,

    username TEXT UNIQUE,

    password TEXT

)

""")

conn.commit()

# ----- AUTH FUNCTIONS -----

def signup_user(username, password):

    username = username.strip()

    password = password.strip()

    if username == "" or password == "":

        return "empty"

try:

    cur.execute("INSERT INTO users (username, password) VALUES (?, ?)",
```

```

        (username, password))

    conn.commit()

    return "success"

except Exception as e:

    print("Signup error:", e)

    return "exists"

def login_user(username, password):

    username = username.strip()

    password = password.strip()

    cur.execute("SELECT * FROM users WHERE username=? AND password=?",

        (username, password))


    result = cur.fetchone()

    print("DEBUG LOGIN:", result) # for checking


    return result


# ----- PRODUCTS -----

products = {

    "Kitchen": [

        {"name": "Cooker", "price": 2500},

        {"name": "Pan", "price": 800}

    ],

    "Makeup": [

        {"name": "Lipstick", "price": 300},

        {"name": "Foundation", "price": 800}

    ],

    "Gadgets": [

```

```

        {"name": "Mobile", "price": 15000},
        {"name": "Headphones", "price": 2000}
    ]
}

cart = {}
current_user = ""
# ----- CART -----

def add_to_cart(product):
    name = product["name"]
    if name in cart:
        cart[name]["qty"] += 1
    else:
        cart[name] = {"price": product["price"], "qty": 1}

    update_cart()

def update_cart():
    cart_listbox.delete(0, tk.END)
    total = 0
    for item, d in cart.items():
        cart_listbox.insert(
            tk.END,
            f'{item} x {d["qty"]} = ₹ {d["price"] * d["qty"]}'
        )
        total += d['price'] * d['qty']

    total_label.config(text=f'Total: ₹ {total}')

def place_order():
    if not cart:

```

```

        messagebox.showwarning("Warning", "Cart empty!")

    return

total = sum(d['price'] * d['qty'] for d in cart.values())

days = random.randint(2, 5)

delivery = datetime.datetime.now() + datetime.timedelta(days=days)
bill = f'Order placed for {current_user}\n\n'
    for item, d in cart.items():
        bill += f'{item} x {d['qty']} = ₹{d['price'] * d['qty']}\n'
    bill += f'\nTotal: ₹{total}'

    bill += f'\nDelivery: {delivery.strftime('%d-%m-%Y')}'

messagebox.showinfo("Order", bill)

cart.clear()

update_cart()

def signup():

    name = entry.get()

    password = password_entry.get()

result = signup_user(name, password)

if result == "success":

    messagebox.showinfo("Success", "Signup successful! Now login")

elif result == "exists":

    messagebox.showerror("Error", "User already exists")

else:

    messagebox.showwarning("Error", "Fill all fields")

def login():

    global current_user

    name = entry.get()

    password = password_entry.get()

    user = login_user(name, password)

```

```

if user:
    current_user = name
    login_frame.pack_forget()
    show_app()
else:
    messagebox.showerror("Error", "Wrong username or password")

# ----- MAIN APP -----

def show_app():
    global cart_listbox, total_label
    frame = tk.Frame(root, bg="#f0f8ff")
    frame.pack(fill="both", expand=True)
    tk.Label(frame,
             text=f"Welcome {current_user} ",
             font=("Arial", 16, "bold"),
             bg="#f0f8ff").pack(pady=10)

# PRODUCTS

for cat, items in products.items():
    tk.Label(frame,
             text=cat,
             font=("Arial", 14, "bold"),
             bg="#f0f8ff").pack()

for p in items:
    row = tk.Frame(frame, bg="#f0f8ff")
    row.pack()
    tk.Label(row,
             text=f"{p['name']} - ₹ {p['price']}",
             width=25,
             anchor="w",
             bg="#f0f8ff").pack(side=tk.LEFT)

```

```

tk.Button(row, text="+",
          command=lambda p=p: add_to_cart(p),
          bg="green", fg="white").pack(side=tk.RIGHT)

# CART

tk.Label(frame,
        text="Cart",
        font=("Arial", 14, "bold"),
        bg="#f0f8ff").pack(pady=10)

cart_listbox = tk.Listbox(frame, width=40)
cart_listbox.pack()

total_label = tk.Label(frame,
                      text="Total: ₹0",
                      font=("Arial", 12, "bold"),
                      bg="#f0f8ff")
total_label.pack()

tk.Button(frame,
        text="Place Order",
        command=place_order,
        bg="blue",
        fg="white").pack(pady=10)

# ----- LOGIN UI -----

root = tk.Tk()
root.title("Shopping App (Fixed Version)")
root.geometry("420x550")

```

```
login_frame = tk.Frame(root)
login_frame.pack(expand=True)

tk.Label(login_frame,
        text="Login / Signup",
        font=("Arial", 16, "bold")).pack(pady=10)

tk.Label(login_frame, text="Username").pack()
entry = tk.Entry(login_frame)
entry.pack()

tk.Label(login_frame, text="Password").pack()
password_entry = tk.Entry(login_frame, show="*")
password_entry.pack()

tk.Button(login_frame,
        text="Login",
        bg="blue",
        fg="white",
        command=login).pack(pady=5)

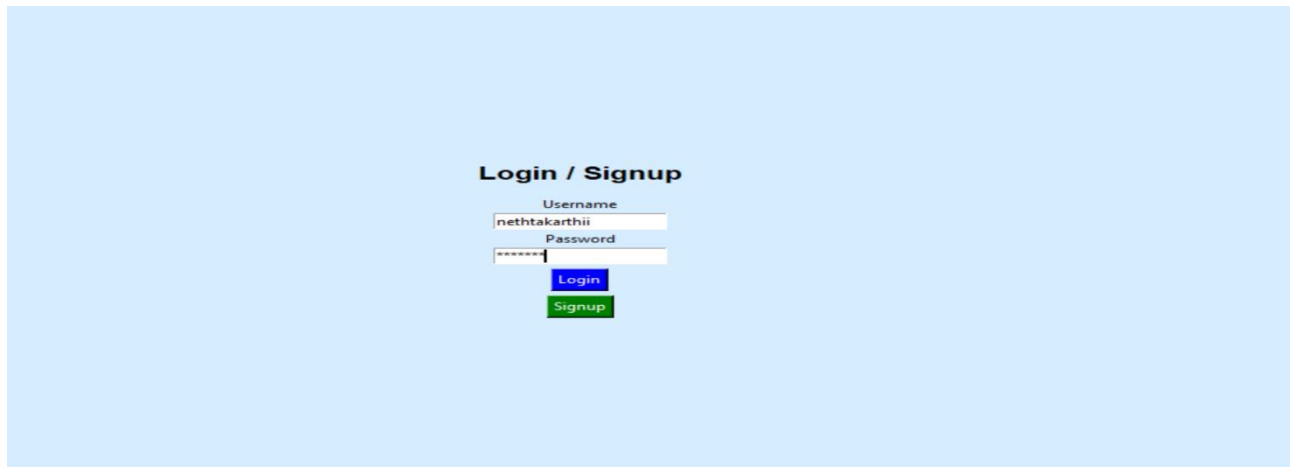
tk.Button(login_frame,
        text="Signup",
        bg="green",
        fg="white",
        command=signup).pack()

root.mainloop()
```

CHAPTER 6

SNAPSHOTS

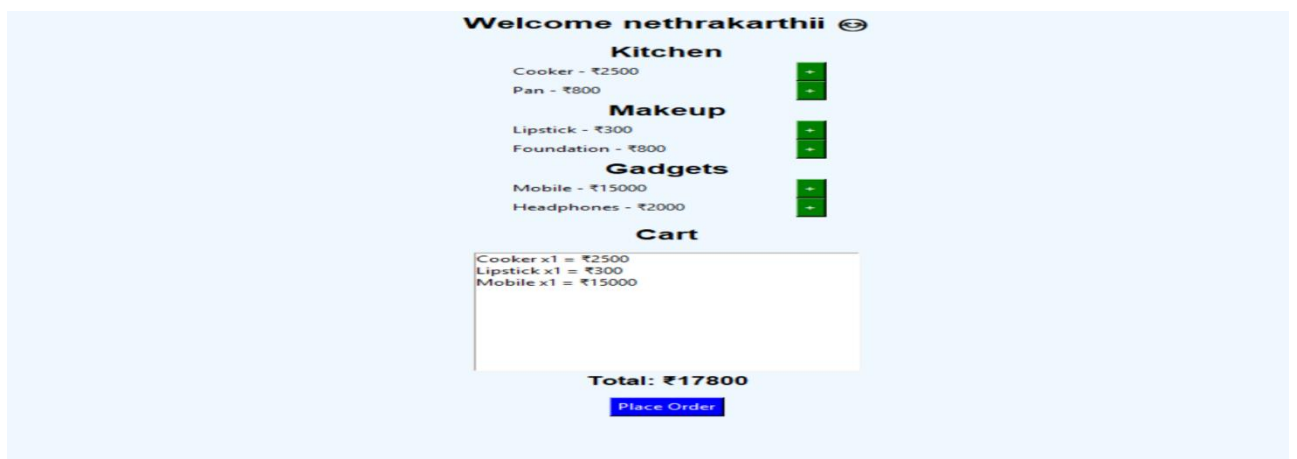
6.1 User login page



The image shows a login/signup page with a light blue background. The title "Login / Signup" is centered at the top. Below it, there are two input fields: "Username" with the text "neththakarthii" and "Password" with masked characters "*****". To the right of the password field are two buttons: a blue "Login" button and a green "Signup" button.

Fig 6.1 Login Page of Online Shopping Cart Manager

6.2 Product Catalog and Cart Management



The image shows a product catalog and cart management page. At the top, it says "Welcome nethrakarthii ☺". Below this, there are three categories: "Kitchen", "Makeup", and "Gadgets". Each category has a list of products with their prices and a green "+" button to add them to the cart. The "Kitchen" category includes "Cooker - ₹2500" and "Pan - ₹800". The "Makeup" category includes "Lipstick - ₹300" and "Foundation - ₹800". The "Gadgets" category includes "Mobile - ₹15000" and "Headphones - ₹2000". Below the product lists is a "Cart" section with a table showing the items added: "Cooker x1 = ₹2500", "Lipstick x1 = ₹300", and "Mobile x1 = ₹15000". At the bottom, it shows the "Total: ₹17800" and a blue "Place Order" button.

Fig 6.2 Product Catalog and Cart Management

6.3 Order Confirmation and Summary

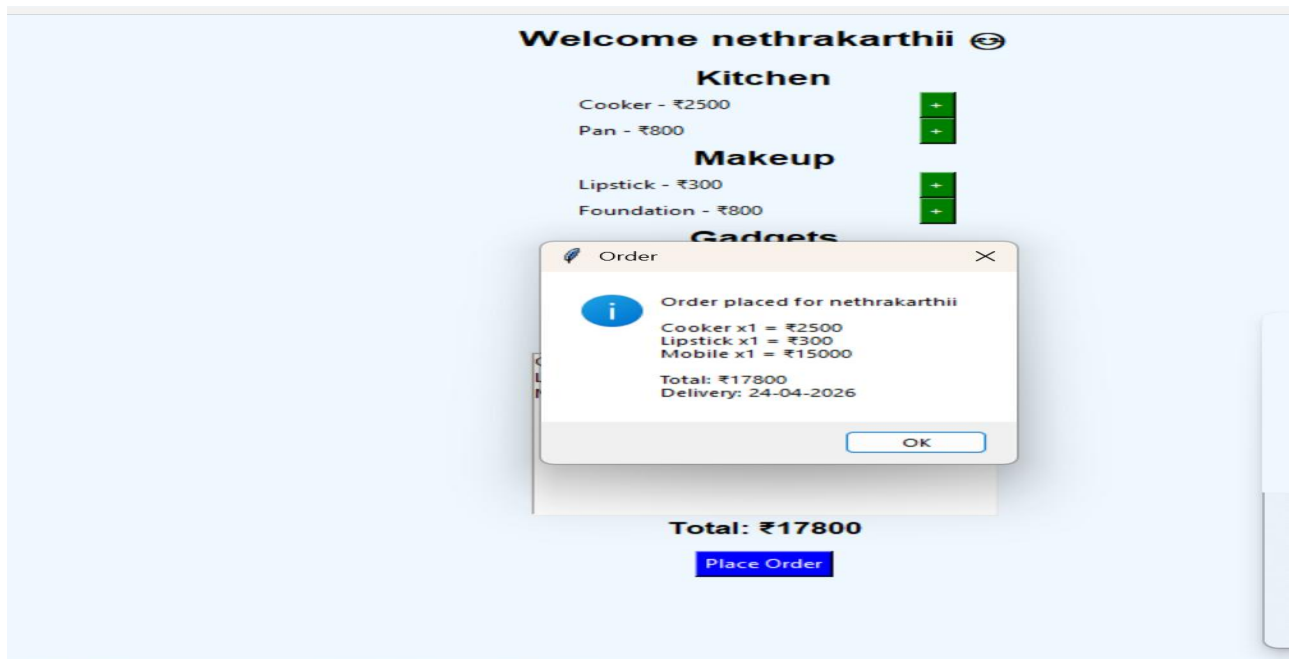


Fig 6.3 Order Confirmation and Summary

CONCLUSION

The Online Shopping Cart Manager is developed to provide a simple, secure, and efficient platform for online shopping activities. The system enables customers to browse products, add items to the shopping cart, manage product quantities, and complete purchases easily through a user-friendly interface. It also helps administrators maintain product details, billing information, and customer order records effectively. The project minimizes manual work, reduces errors in order management, improves data accuracy, and saves time for both users and administrators. By integrating modules such as product management, cart management, billing, and display management, the system enhances the overall shopping experience and provides better convenience, reliability, and efficiency in online purchasing.

REFERENCES

1. Lutz, Mark. *Learning Python*. O'Reilly Media.
2. Sweigart, Al. *Automate the Boring Stuff with Python*. No Starch Press.
3. Python Official Documentation – <https://docs.python.org/3/>
4. GeeksforGeeks – Python Programming Language
<https://www.geeksforgeeks.org/python-programming-language/>
5. Programming with Mosh – Python Full Course
https://www.youtube.com/watch?v=_uQrJ0TkZlc
6. freeCodeCamp.org – Python Tkinter Tutorial
<https://www.youtube.com/watch?v=YXPyB4XeYLA>